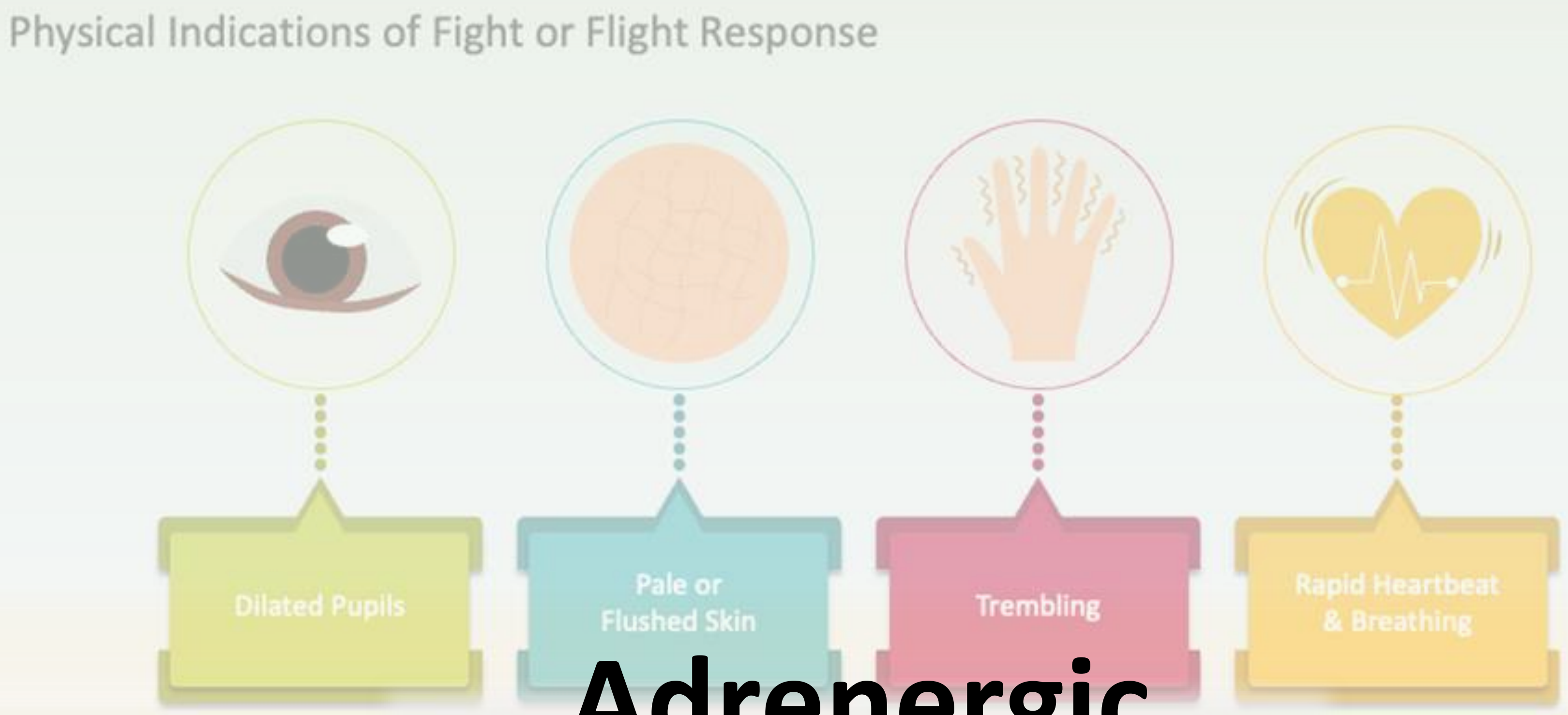

Adrenergic Receptors

The α and β receptor functions, agonists, antagonists, and major clinical uses

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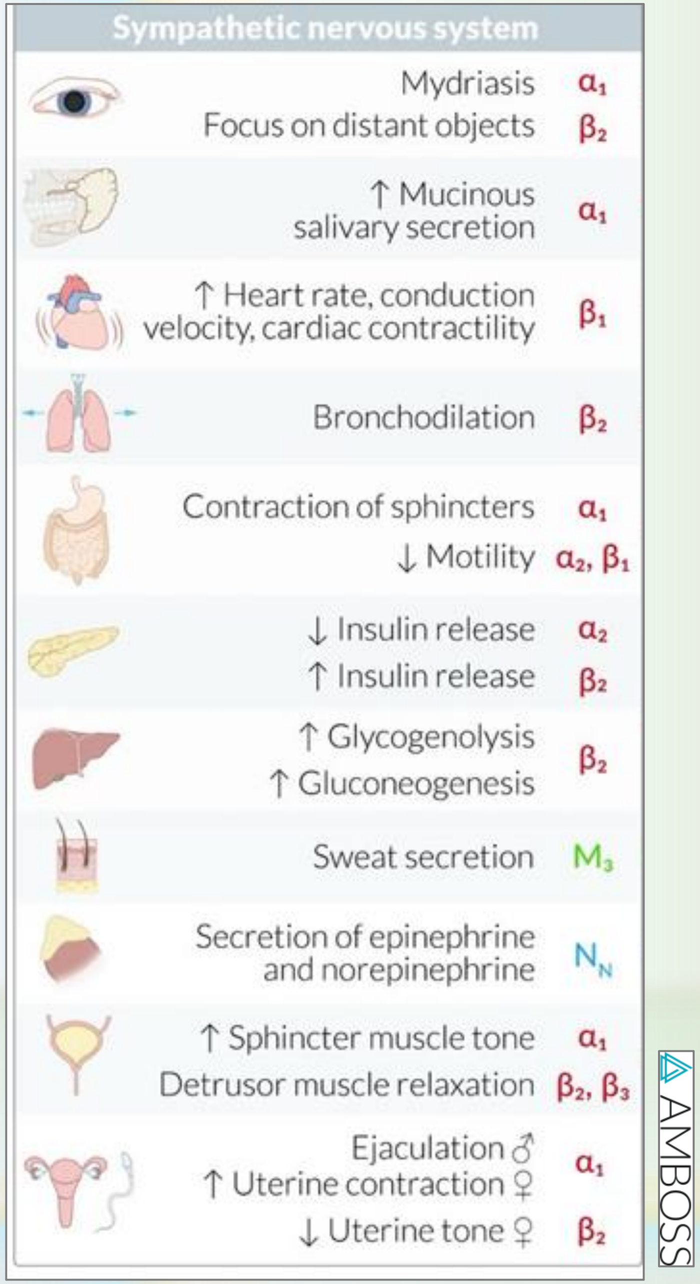
Learning Objectives

By the end of this lecture, you should be able to:

1. Differentiate **α** and **β** adrenergic receptor subtypes.
2. Predict **physiological effects** to simulation/blockade.
3. Compare selective vs non-selective adrenergic **blockers**.
4. Apply receptor knowledge to **drugs** selection.
5. Identify key **adverse effects** and contraindications.



Why Sympathetic Nervous System Matters ??

What happens to heart rate in fight or flight?

Sympathetic activation = fight or flight

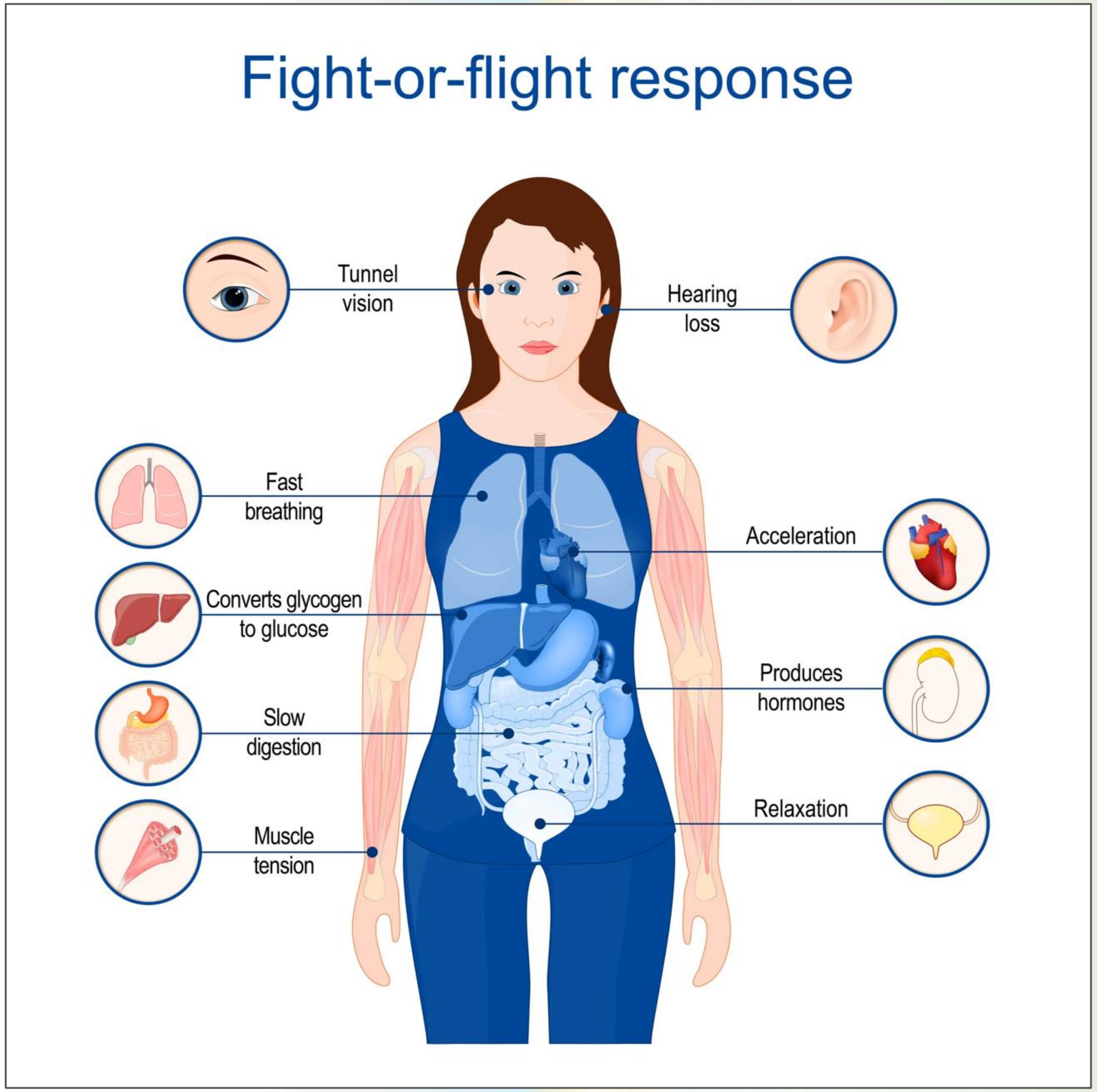
1. ↑ Heart rate
2. ↑ Blood pressure
3. Bronchodilation
4. ↑ Glucose release

- All mediated via **α** and **β receptors** (β_1 = heart, α_1 = vessels, β_2 = lungs)
- Mediated by **NE** (sympathetic nerves) and **epinephrine** (adrenal medulla)

Sympathetic Nervous System

1. Prepares the body for stress
2. Neurotransmitter: Norepinephrine (mostly)
3. Receptors: **α** and **β**
4. Different organ effects depend on receptor subtype distribution

NE: primary neurotransmitter
Epi: hormone circulation



https://www.simplypsychology.org/fight-flight-freeze-fawn.html

Receptors Location

Division	Pre	Post	Target	Clinical Relevance
Parasympathetic	ACh → N.G	ACh → **M2**	Heart	↓ HR (bradycardia). Atropine blocks this → ↑ HR
		ACh → **M3**	Endothelium (limited)	ACh causes NO-mediated vasodilation (seen in pharmacologic testing)
Sympathetic (Adrenergic)	ACh → N.G	NE → $\boldsymbol{\alpha_1}$	• Vessels (vessels, splanchnic) • Eye radial muscles	• Vasoconstriction → ↑ BP (shock, stress) • Radial=dilator → Mydriasis (enlarged pupil)
		NE → $\boldsymbol{\beta_1}$	Heart	↑ HR & contractility (heart failure drugs target this)
		NE → $\boldsymbol{\beta_2}$	• Skeletal muscles vessels • Uterus • Bronchi	Vasodilation (β_2 predominates in skeletal muscle vessels)

N.G: Nicotinic ganglia. **M:** Muscarinic receptor. **ACh:** Acetylcholine. **NE:** Norepinephrine

Receptors Location

Division	Pre	Post	Target	Clinical Relevance
Sympathetic (Cholinergic)	ACh → N.G	ACh → M3	Sweat glands	Sweating. Blocked by anticholinergics (dry skin)
Sympathetic (Dopaminergic)	ACh → N.G	D → D1	Renal vessels	Low-dose dopamine → renal vasodilation, they used to think it protect the kidneys (Not correct) → Not clinically recommended for renal protection (Correct)
Adrenal Medulla	ACh → N.A	EPI/NE (blood)	Heart & vessels	Fight-or-flight systemic response

N.G: Nicotinic ganglia. **M:** Muscarinic receptor. **ACh:** Acetylcholine. **NE:** Norepinephrine. **EPI:** Epinephrine. **D:** Dopamine. **D1:** Dopamine receptor

Adrenergic Receptor Types

β_1 = stimulation, **α_1** = constriction, **β_2** = relaxation

Receptor	Coupling	Main Location	Main Effect
α_1	Gq	Vessels, bladder	Contraction of smooth muscles
α_2	Gi	Presynaptic	↓ NE release
β_1	Gs	Heart	↑ HR, ↑ contractility (stimulate the heart)
β_2		Lungs, vessels	Relaxation of smooth muscles
β_3		Adipose tissues (mainly) Urinary bladder	Lipolysis and Thermogenesis in brown fat. Relaxation of the bladder and prevention of urination.

Gq: Stimulatory. **Gs**: Stimulatory. **Gi:** Inhibitory G-protein

Adrenergic Neurotransmitters

Drug	Dose-dependent effect			Clinical Pearl
	Low	Moderate	High	
Epinephrine	β_2 (vasodilation)	β_1	α_1	Used in anaphylaxis because it gives: Bronchodilation (β_2) + vasoconstriction (α_1) + ↑ cardiac output (β_1)
Epinephrine: "dose-dependent switch" ($\beta \rightarrow \alpha$), meaning it acts on α and β receptors (broader activation)				
Norepinephrine	α_1 dominant	$\alpha_1 + \beta_1$	α_1↑	Drug of choice in septic shock: Strong vasoconstriction increases BP (+ reflex bradycardia)
Norepinephrine: "always α-dominant" (no real switch). Meaning it mainly acts on α and β_1, **less on** β2 (mainly vessels + heart)				
Dopamine	D_1	β_1	α_1	**D_1:** renal vasodilation. May increase renal perfusion, but it is clinically unreliable (NOT recommended for renal protection, may cause arythmias) **β_1 :** cardiac support **α_1 :** norepinephrine-like effect
Dopamine: has a clear receptor progression with increasing dose. Meaning it is dose-dependent				

Adrenergic Heart Rate Changes

(Baroreceptor Reflex)

↓ Blood Pressure → ↑ Heart Rate (Reflex Tachycardia)
↑ Blood Pressure → ↓ Heart Rate (Reflex Bradycardia)

1. α_1 Agonist (e.g., Phenylephrine)

↑ Vasoconstriction → ↑ BP → Reflex ↓ HR (Bradycardia)

2. α_1 Blocker (e.g., Terazosin)

↑ Vasodilation → ↓ BP → Reflex ↑ HR (Tachycardia)

3. β_1 Agonist (e.g., Isoproterenol)

Direct ↑ HR. Direct ↑ contractility. ↑ conduction

Reflex tachycardia is most seen with vasodilators (↓ BP)
* Reflex is mediated via baroreceptors

ANTAGONIST = BLOCKERS = SYMPATHOLYTIC

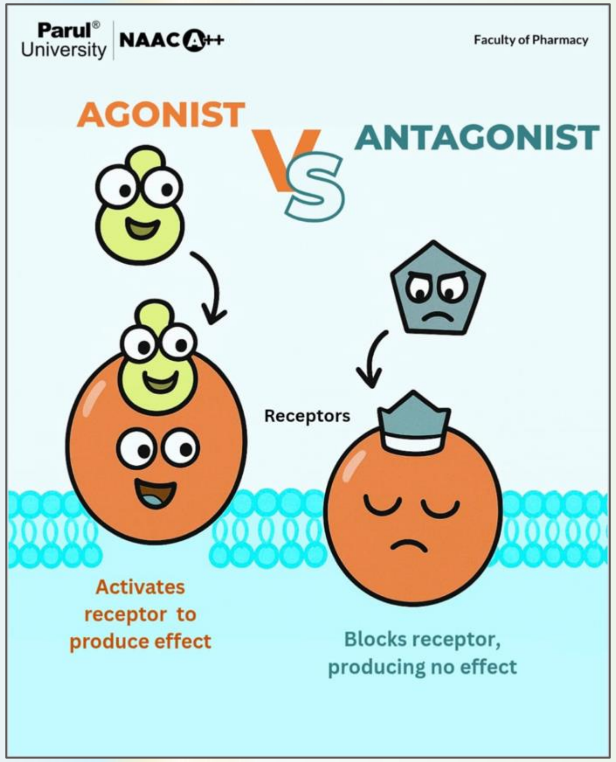
Adrenergic receptor Antagonists:

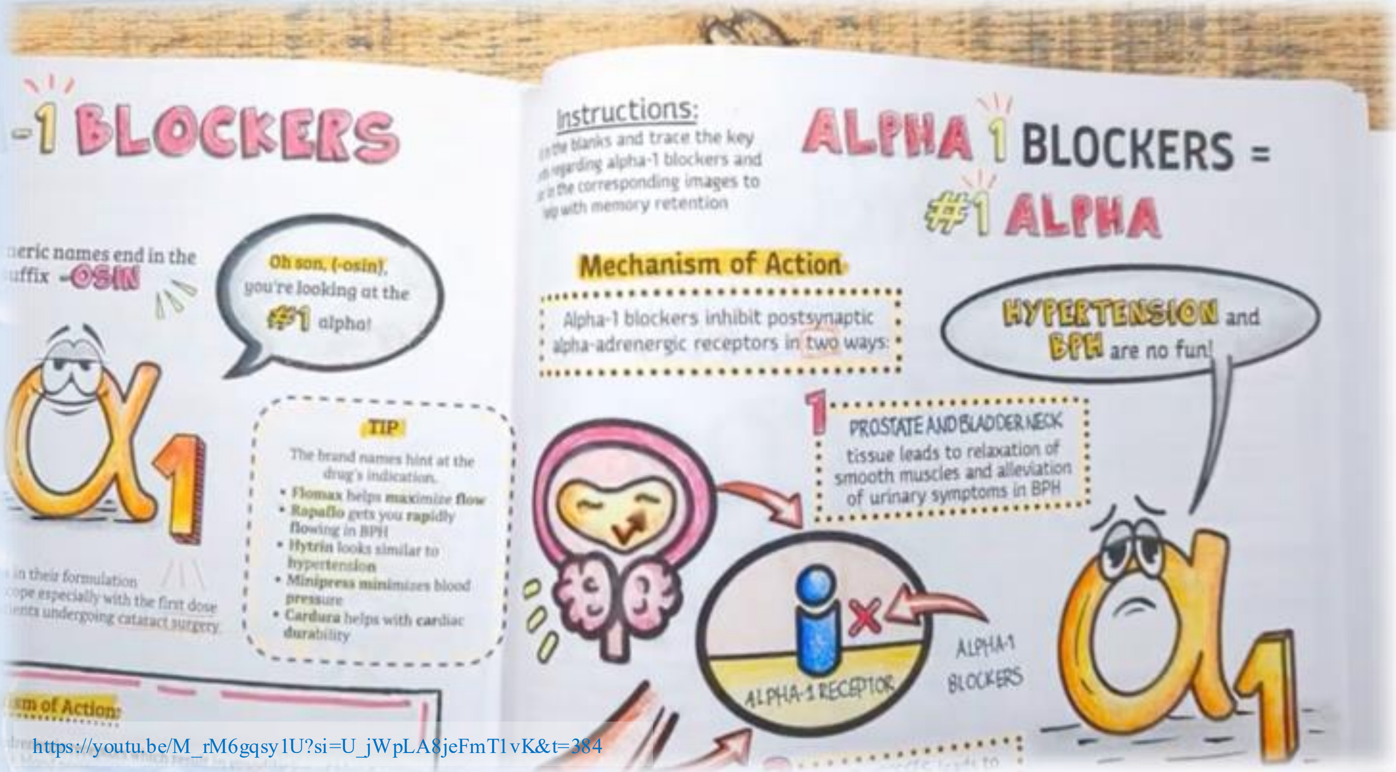
- **Adrenergic agonists** mimic sympathetic activation
- **Adrenergic antagonists** block sympathetic effects

➢ **Block α** or **β adrenergic receptors → Prevent** effects of Epi/NE

Adrenergic Receptor Antagonists Classification:

1. α blockers
2. β blockers
3. Mixed (α + β) blockers





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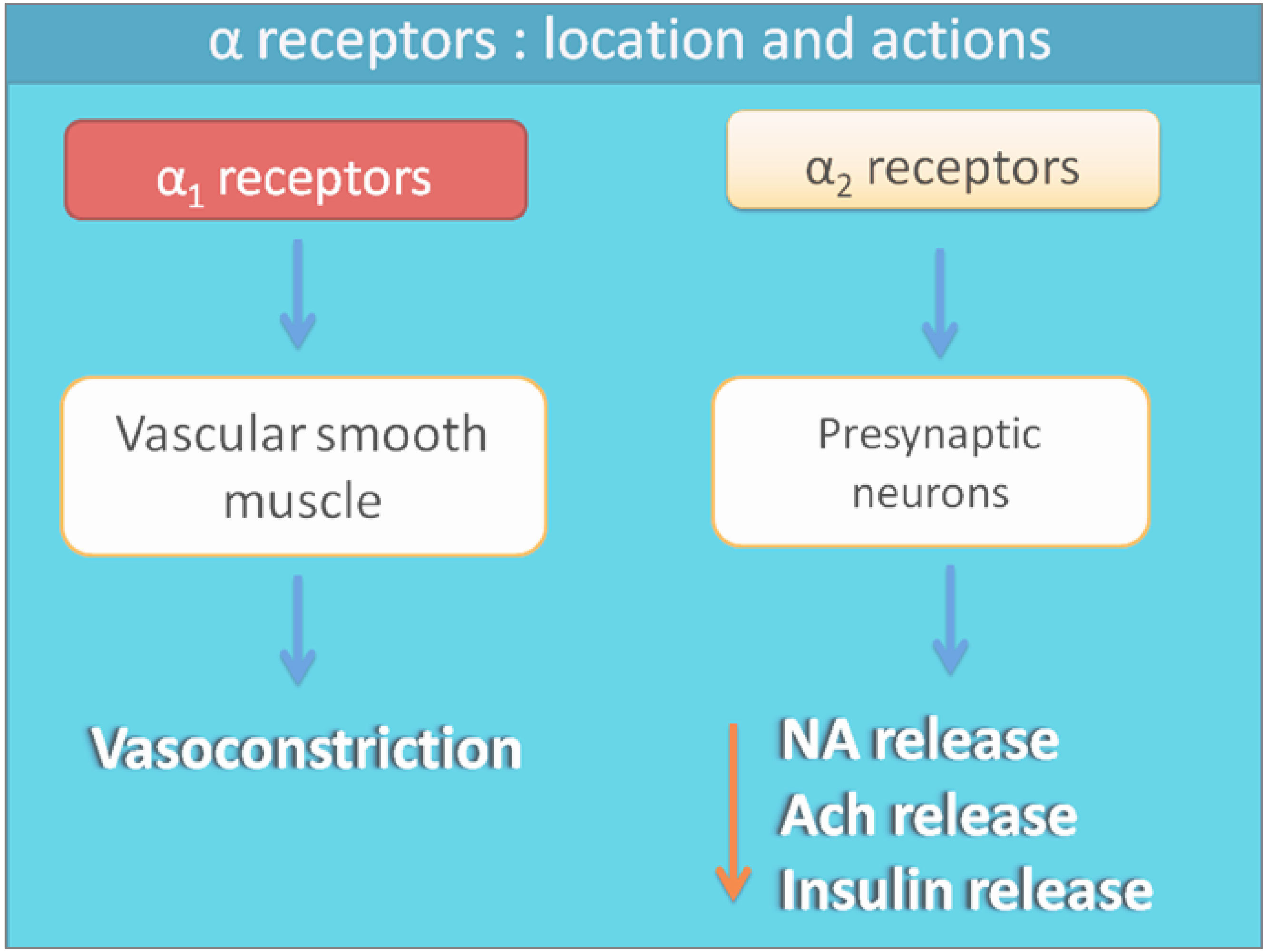
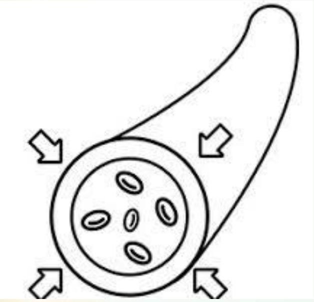


Adrenergic α-Receptors

Receptor	Agonist	Antagonist
α_1	Phenylephrine	Prazosin
α_2	Clonidine	Yohimbine
$\alpha_1 + \alpha_2$	Oxymetazoline	Phentolamine

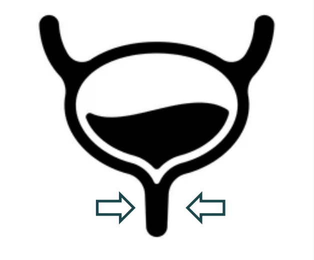
α_1 Receptor Agonists

1. The receptor is coupled to the stimulatory **Gq**-protein subunit that targets phospholipase C (PLC) and result in increase in calcium levels.
2. Ca^{2+} Causes smooth muscle contraction



Clinical effects:

1. Vasoconstriction → ↑ peripheral resistance → ↑ BP
2. Urinary sphincter contraction/prostate contraction
3. Mydriasis



α_1 = squeeze !!
α_1 activation = vasoconstriction → ↑ BP
α_1 blockade = vasodilation → first-dose hypotension

α_1 Receptor Agonists

Example: Phenylephrine

1. α1 agonist (sympathomimetic agent)
2. Primarily used as an OTC decongestant for nasal/sinus congestion
3. Used intravenously to treat severe hypotension

Clinical effects:

1. Nasal Decongestant: ↓ congestion (e.g., allergies), by constricting blood vessels in the nasal mucosa.

Note: Nasal spray should not be used for more than 3 days, as overusing it can make congestion worse (**Rebound Congestion**)

2. Hypotension: It increase both systolic and diastolic blood pressure.

Note: This increase in pressure can lead to reflex bradycardia

3. Ophthalmic Agent: eye drops to dilate the pupils (mydriasis) for eye examinations

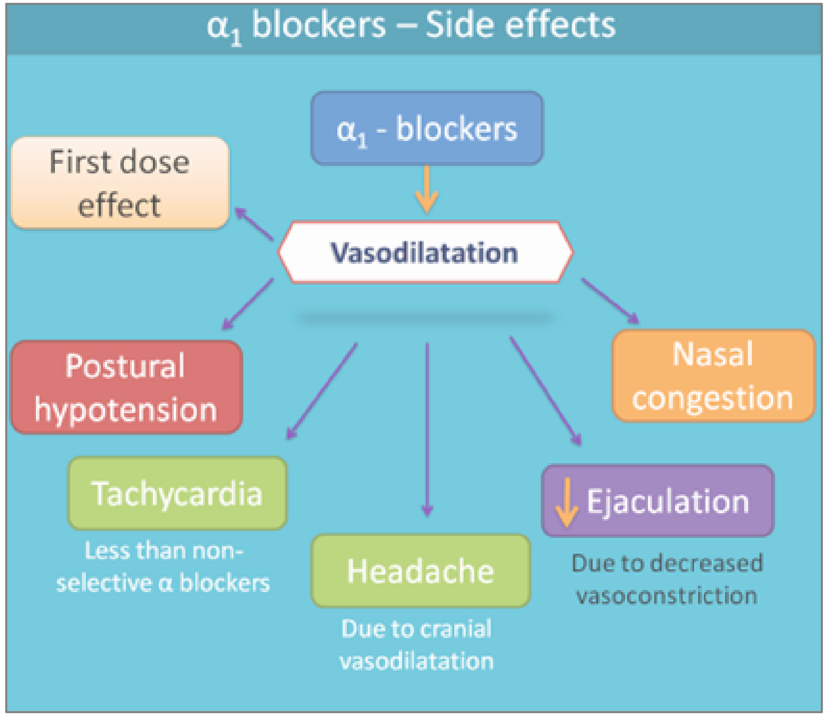
α_1 Receptor Antagonists

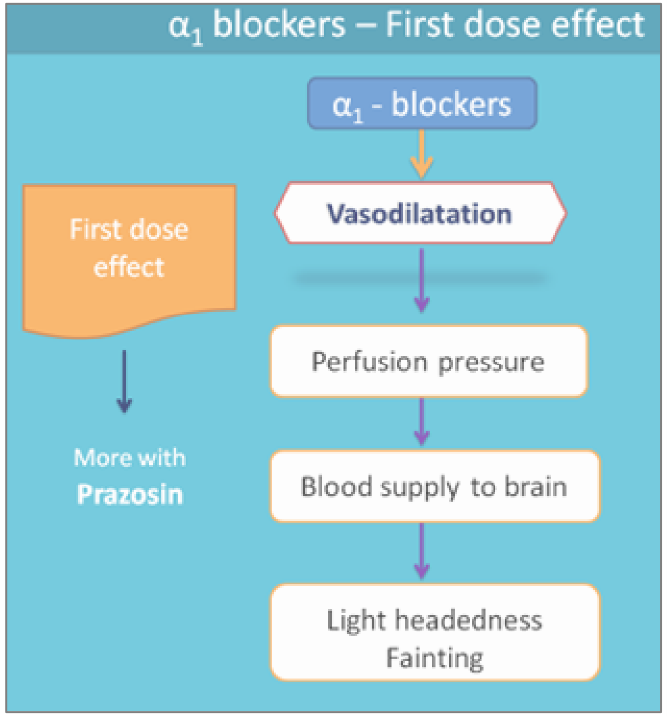
Examples: Praz**osin**, Doxaz**osin**, Tamsul**osin**

Drug	Receptor	Location	Effect	Main Use
Prazosin	α_{1B}	Blood vessels	1. Causes relaxation of arterial and venous smooth muscle 2. Decrease peripheral vascular resistance 3. Lower blood pressure with less reflex tachycardia (because no α2)	Hypertension
Tamsulosin	α_{1A}	Prostate and bladder	Causes relaxation in the smooth muscle of the bladder neck and prostate and improves urine flow with less pronounced effects on blood pressure	Benign Prostatic Hyperplasia (BPH)

Tamsulosin = increase urine flow

α_1 Receptor Antagonists Adverse





First-dose Phenomenon

First-dose syncope → "First dose, first faint

Is classic for **Prazosin**. Advise patient to take first dose at bedtime.

α_2 Receptor Agonists

Example: Clonidine

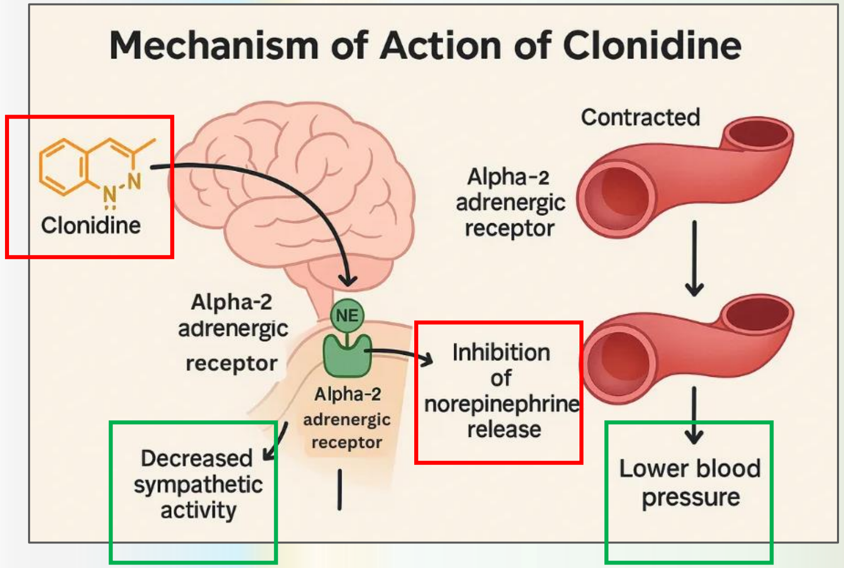
- This receptor is coupled to the inhibitory **Gi**-protein subunit that targets cAMP and result in inhibition of NE release (presynaptic).
- **Actions:** reduced NE release, reduce sympathetic outflow slow your nervous system to keep everything calm

Clinical effects:

Central effects → reduces sympathetic outflow →BP reduction

Clonidine (**α_2 agonist)** → lowers BP centrally

α_2 Receptors are presynaptic and inhibit norepinephrine release.
Clonidine lowers BP by reducing sympathetic outflow



Sudden withdrawal →
rebound hypertension (life-threatening)

α_2 Receptor Antagonists

Example: Yohimbine, Idazoxan, Mirtazapine

These antagonists inhibit the negative feedback of norepinephrine, thereby stimulating the sympathetic nervous system. However, there is limited evidence on the clinical significance of this mechanism in human medicine

Idazoxan: used in research
Yohimbine: historically, used for erectile dysfunction (limited evidence)
Mirtazapine: used primarily to treat depression

α_2 is a pre-synaptic inhibitory receptor:
Blocked α_2 receptor on presynaptic sympathetic nerve:
→ increase N.E release
→ stimulation of β_1
→ increases heart rate and cardiac output

α_1 and α_2 receptors Agonists :
(On Synapse)

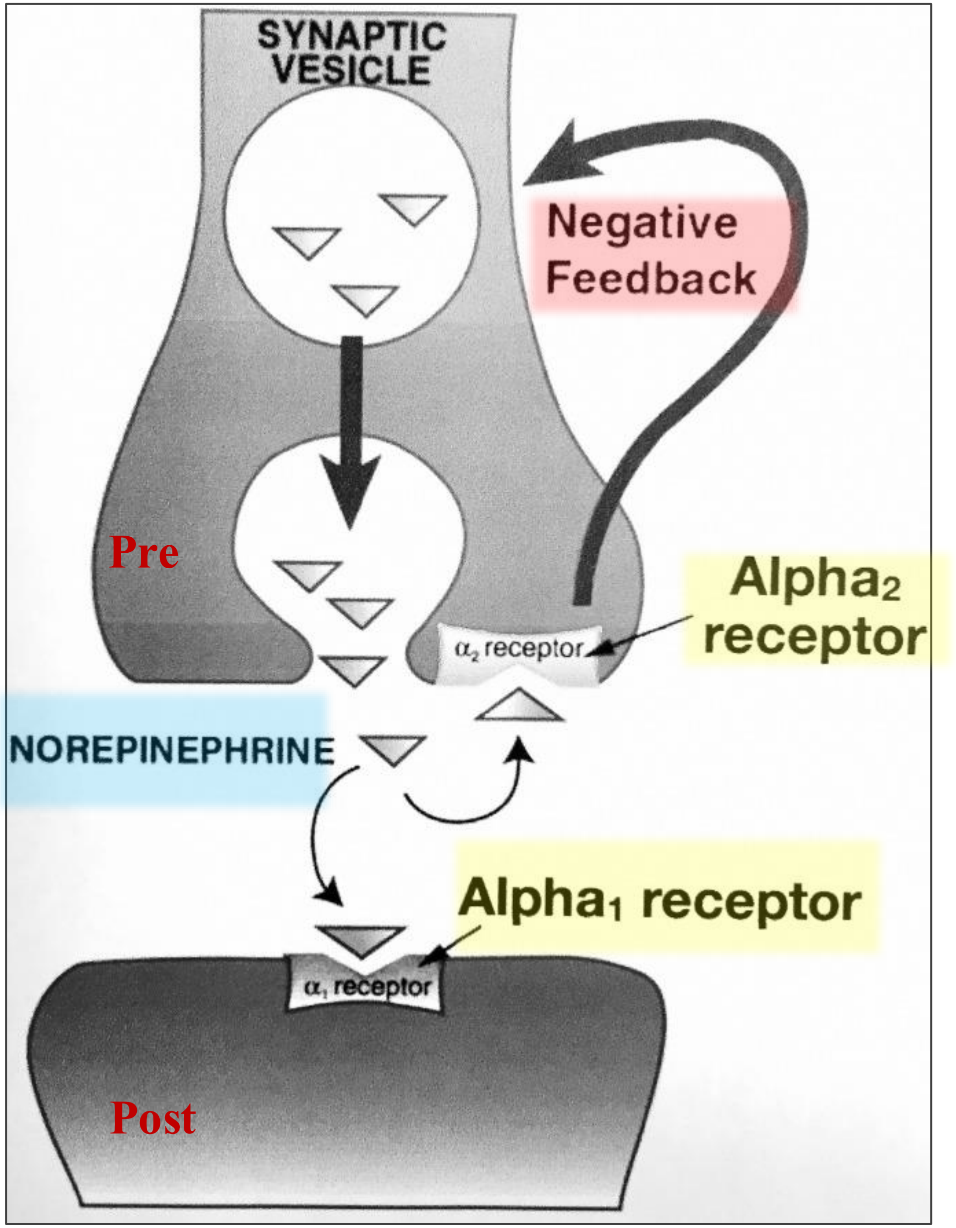
Examples: Oxymetazoline, Phentolamine

Post-junctional α_1 receptor	Pre-junctional $\alpha_2$receptor
Mediate effects on target tissues (α_1 contracts smooth muscles)	Inhibit NE release through negative feedback

Negative feedback loop activation:
Results in a reduction in heart rate and blood pressure, and attenuation of the sympathetic stress response

$\boxed{\alpha 2 = \text{negative feedback} \rightarrow \downarrow \text{NE}}$

Uses:
- Common nasal decongestants
- The non-selectivity of these drugs limits their use in treating nasal congestion or managing intraocular pressure (due to systemic effects)



https://pmc.ncbi.nlm.nih.gov/articles/PMC4389556/

α_1 & α_2 Receptor Antagonists
(Non-selective)

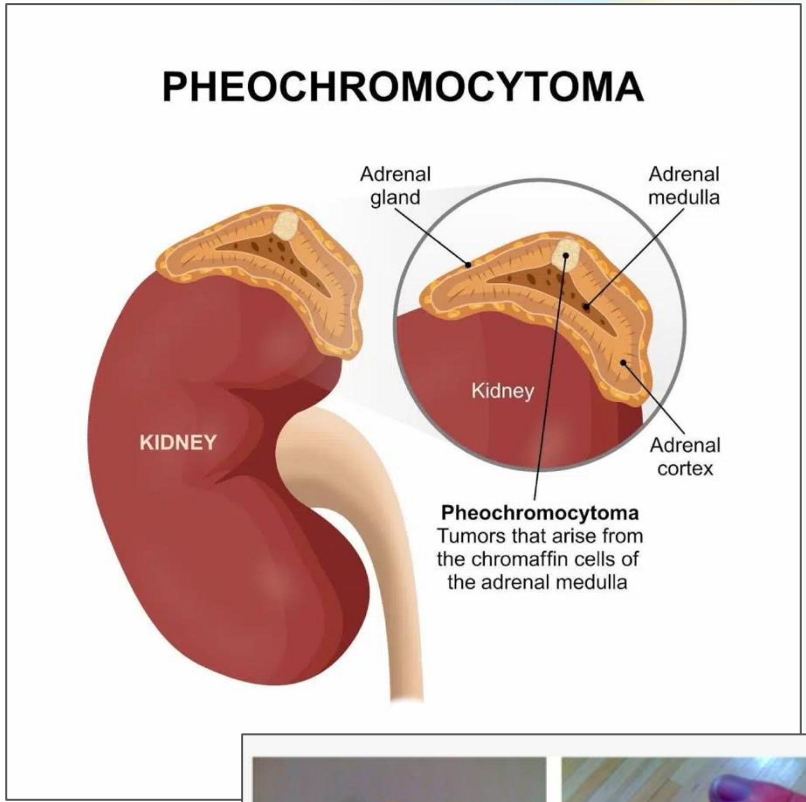
Examples: Phenoxybenzamine, Phentolamine

Phenoxybenzamine Use:

1. **Pheochromocytoma:**
   A catecholamine-secreting tumor
   Epinephrine interacts with both α and β receptors
   Treatment: Phenoxybenzamine + β blocker

2. **Raynaud's disease:**
   Spasms in the small blood vessels of the fingers and toes

- **Phenoxybenzamine** is irreversible, non-selective α blocker
- **Phentolamine** is reversible, non-selective α blocker
- **ALWAYS** give α-blocker **BEFORE** β-blocker
  (Prevents unopposed α → severe hypertension crisis)



High epinephrine:
→ α_1 stimulation
→ Severe vasoconstriction
→ Hypertensive crisis
→ Cerebral hemorrhage

Raynaud's disease: Phenoxybenzamine relief the vasospasm

α_1 & α_2 Receptor Antagonists
(Non-selective)

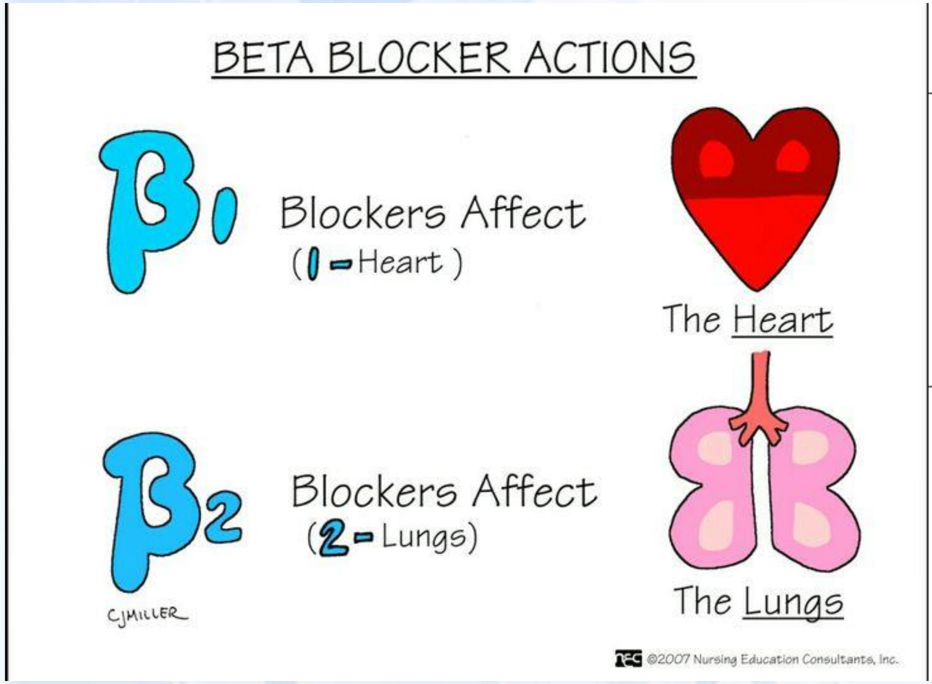
Examples: Phenoxybenzamine, Phentolamine

Phentolamine Use:

1. Pheochromocytoma
2. Hypertensive crises (abrupt withdrawal of clonidine)
3. Local antidote for extravasation of norepinephrine (i.e., vasopressor). Norepinephrine is an α-adrenergic agonist commonly used for the treatment of shock

Subcutaneous tissue ischaemia following extravasation of norepinephrine during intravenous infusion.

Extravasation from a peripheral cannula
Induce subcutaneous tissue ischaemia (through vasoconstriction) and can lead to skin necrosis
Treatment: discontinuation of the intravenous site, pharmacological vasodilators (Phentolamine) and wound care.


BETA BLOCKER ACTIONS
β1 Blockers Affect
(1 - Heart)
The Heart
β2 Blockers Affect
(2 - Lungs)
The Lungs
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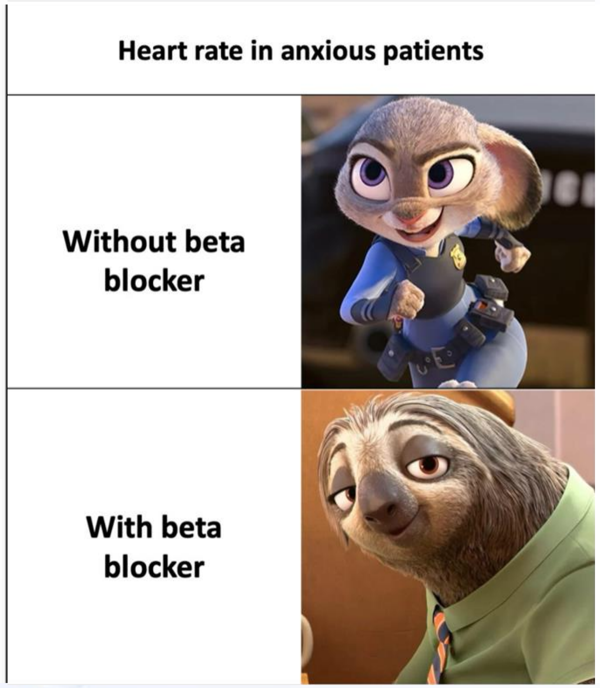
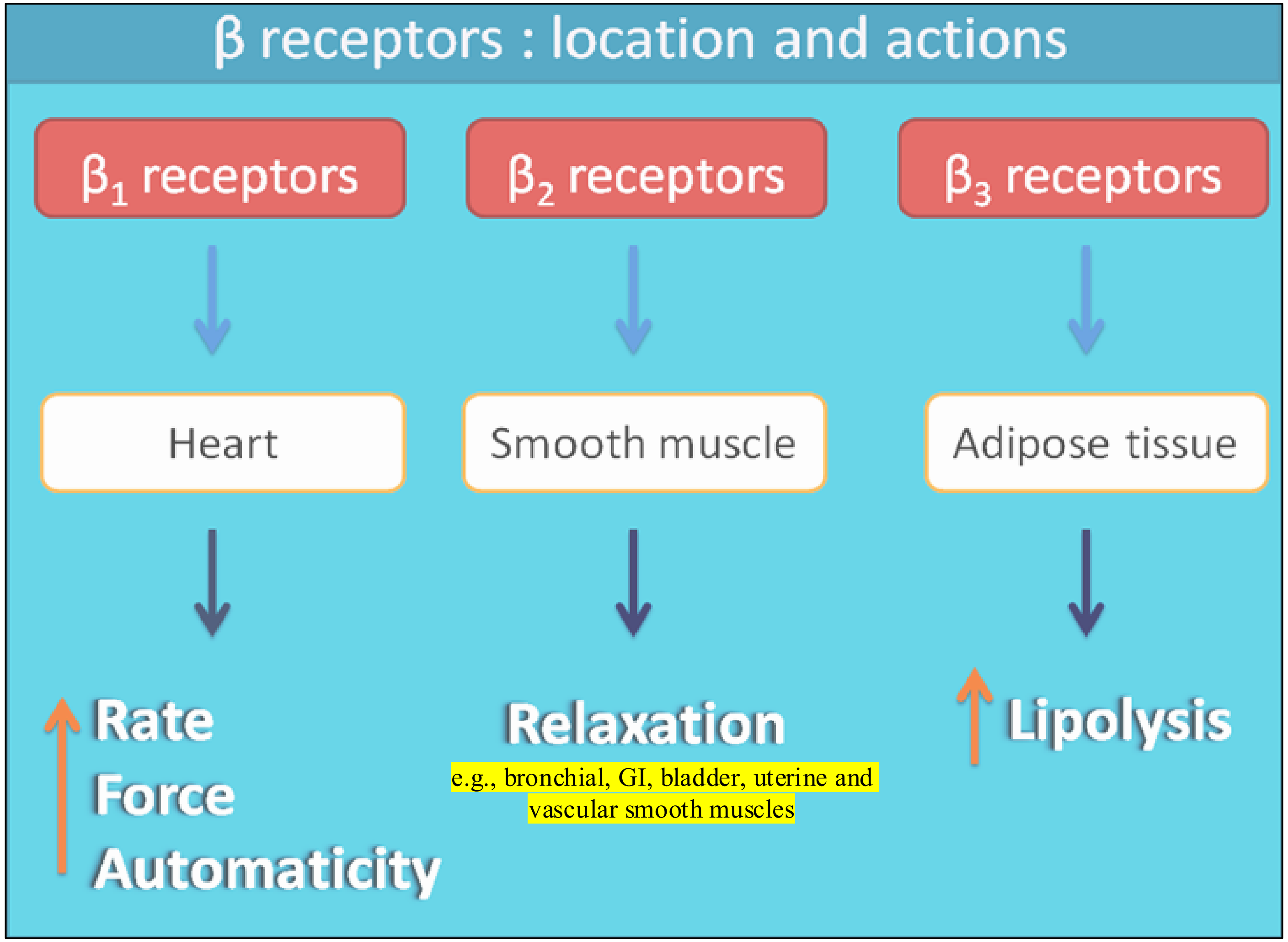

Heart rate in anxious patients
Without beta blocker
With beta blocker


Beta Blocker



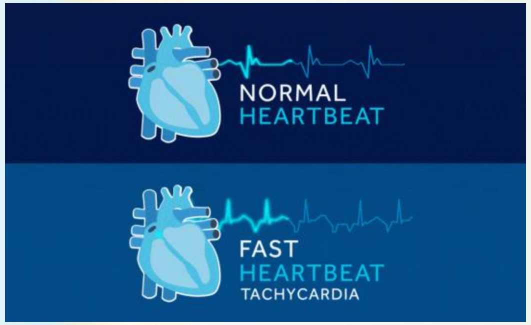
Adrenergic β-Receptors

Receptor	Agonist	Antagonist
β_1	Dobutamine	Metoprolol
β_2	Salbutamol	Research use
β_3	Mirabegron	Research use
$B_1 + \beta_2$	Isoprenaline	Propranolol
$B_1 + \beta_2 + \alpha 1$	Epinephrine	Labetalol

β_1 Receptor Agonists

Example: Dobutamine

1. The receptor is coupled to the stimulatory **Gs**-protein subunit that targets cAMP and result in increase protein phosphorylation.
2. Protein phosphorylation Causes stimulatory effects on the heart

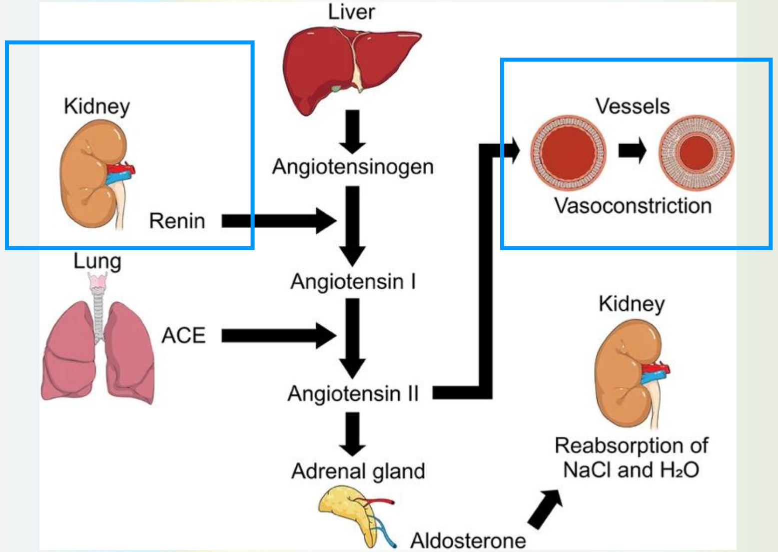


Speed

Intrinsic Strength

Clinical effects:

↑ HR, ↑ contractility, ↑ conduction
↑ Renin release (increases BP)



Dobutamine: mostly β_1 (with some β_2, α_1 actions)
Dobutamine → acts on heart muscles β_1 to increase output in CHF
Dopamine → is better at raising blood pressure (shock, hypotension)

β_1 Receptor Agonists

Under stressful conditions:

- The sympathetic system is activated to increase blood supply to vital organs:

  (Activation of β_1-receptors → ↑ cardiac output → ↑ blood supply → ↑ energy)

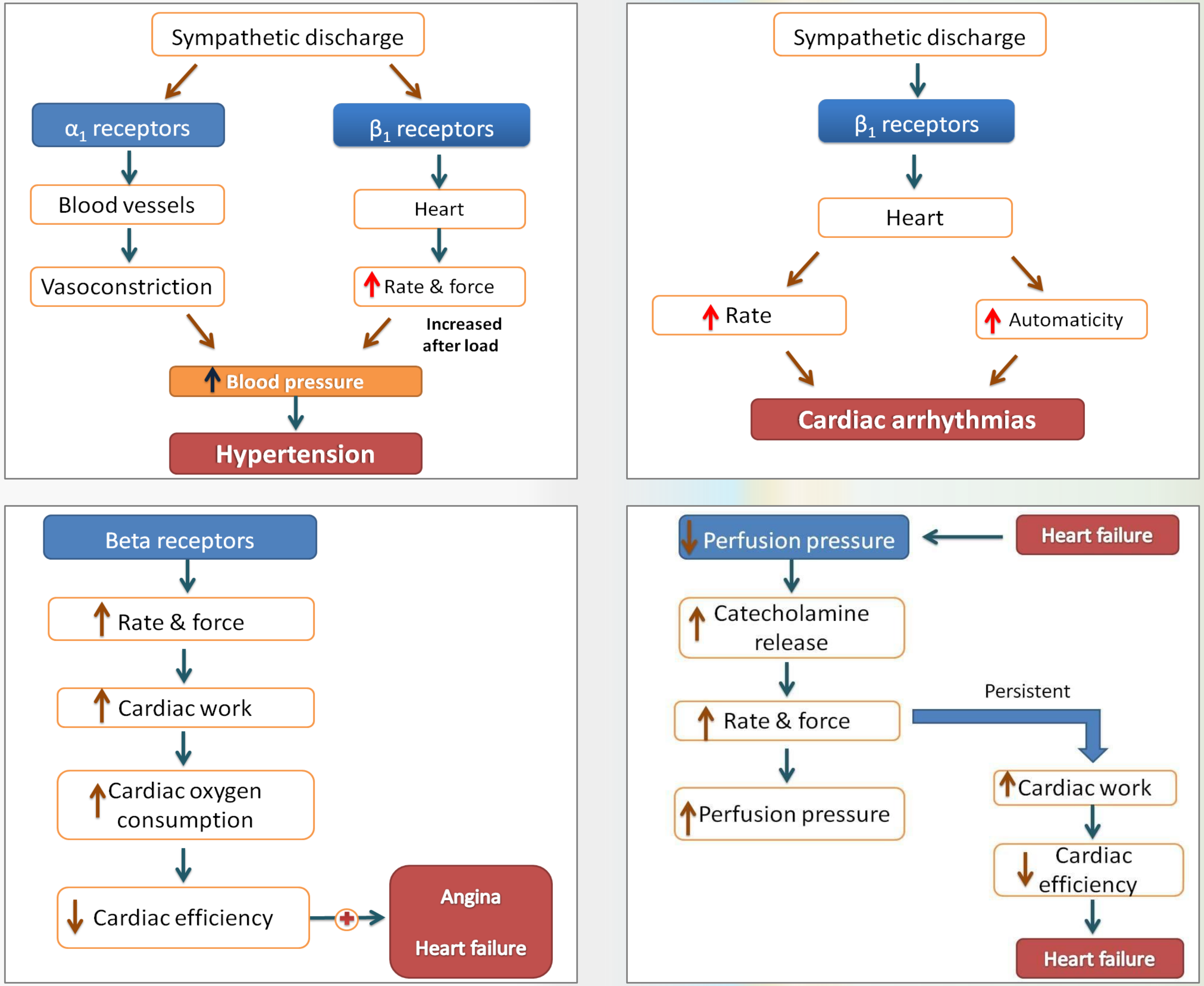
- β_1-receptor activation is **essential**
- Short-term activation of β_1-receptor is physiologic and useful (exercise, stress)
- Persistent sympathetic stimulation can become **pathological**.
- Cardiovascular disorders like HTN, angina, and arrhythmias are associated with **persistent** increase in sympathetic system requiring the use of β_1- blockers

Chronic stimulation → harmful (HF, arrhythmia)

Chronic Catecholamine Elevation

Chronic unregulated stimulation (e.g., Heart failure, persistent catecholamine excess) causes:

1. Remodeling/dilation of the heart
2. Fibrosis/cell death
3. Reduced cardiac functions



β_1 Receptor Antagonists

Examples: Metoprolol, Atenolol

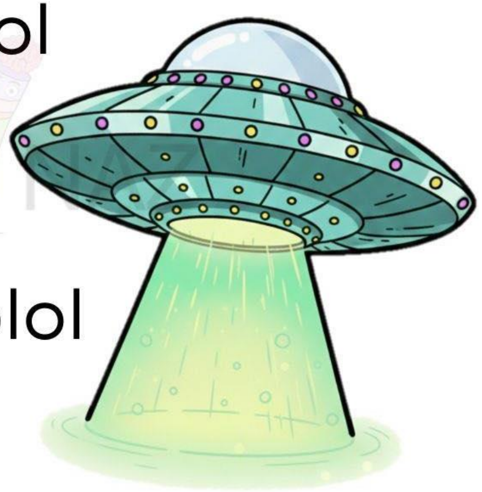
- β_1-receptors are cardio-selective at LOW dose only
- β_1 selectivity is dose-dependent.
- At high doses, even selective β_1-blockers lose selectivity.

Uses:

1. Hypertension
2. Ischemic heart disease
3. Stable heart failure
4. COPD (safer than non-selective)

Mnemonic: " A-BEAM"

A - Acebutolol
B - Bisoprolol
E - Esmolol
A - Atenolol
M- Metoprolol



Cardio-selective

β_1 Receptor Antagonists

Rapid-acting, ultra-short acting (minutes): Esmolol

- Used in critical care and surgical settings to treat tachycardia and high BP
- Very short half-life (~9 min), IV infusion administration
- Rapid metabolism ensures that cardiovascular effects diminish quickly, typically within 10-3 min

Uses:

Acute management of atrial fibrillation, atrial flutter, intraoperative tachycardia and high BP

Ideal for scenarios requiring rapid, temporary control of heart rate and blood pressure

Performance Anxiety

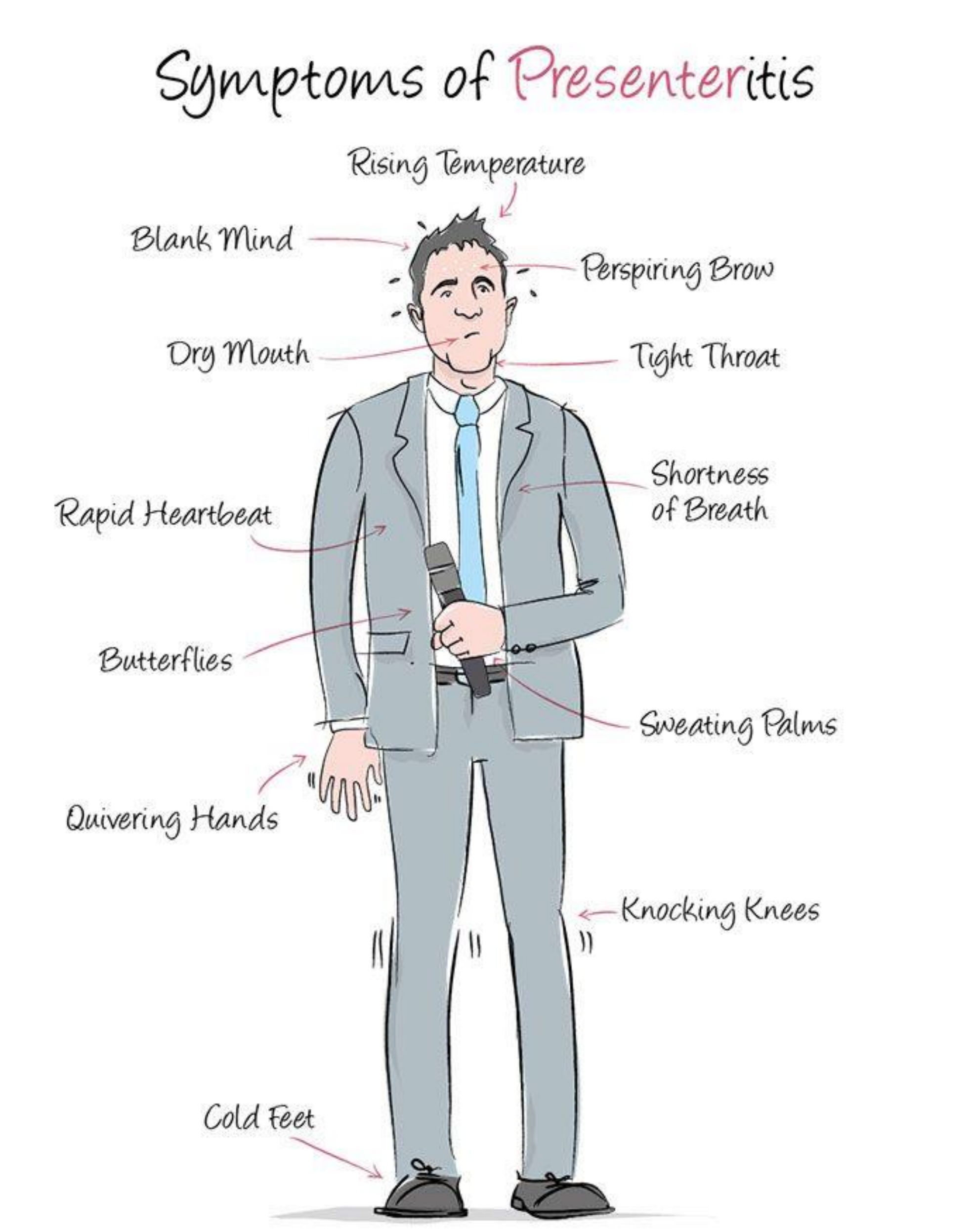
- Stressful situations may trigger fight-or-flight response
- Causes anxious feelings and physical changes like:

  - Increased heart rate
  - Hands, legs and voice shake
  - Sweat
  - Tensed muscles

  https://health.clevelandclinic.org/performance-anxiety-stage-fright

- Adrenaline causes the anxiety symptoms
- Blocking adrenaline lowers the physical symptoms of performance anxiety

  → Performers use **β-blockers**



https://performzen.com/beta-blockers-for-anxiety-guide/

β_2 Receptor Agonists

Example: Salbutamol

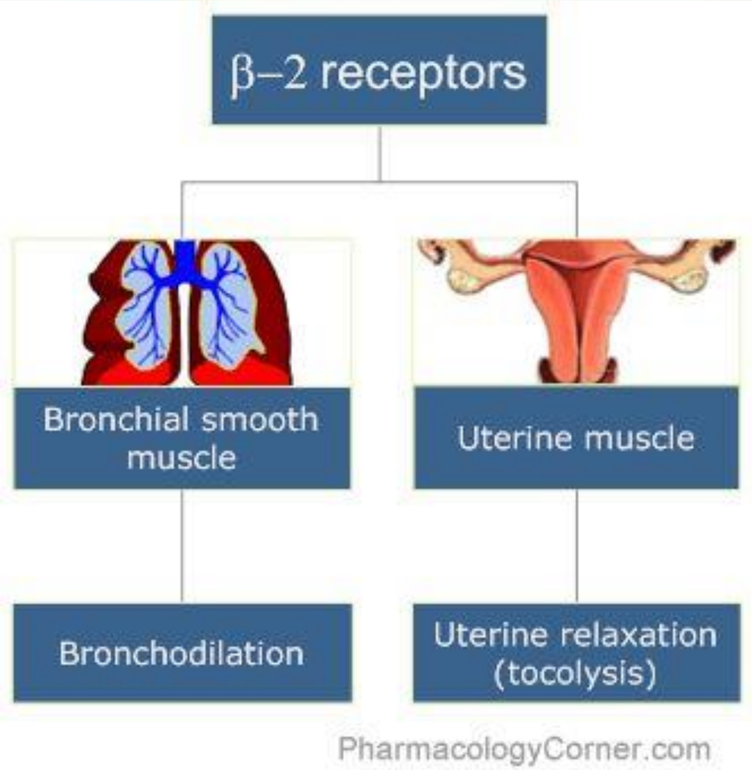
1. The receptor is coupled to the stimulatory **Gs**-protein subunit that targets cAMP and result in increase protein phosphorylation.
2. Results in smooth muscles relaxation (asthma patients)

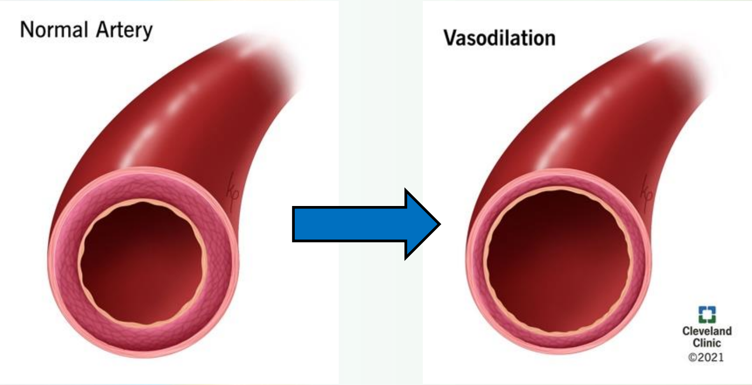
Clinical effects:

1. Bronchodilation (β_2 agonist → asthma treatment)
2. Vasodilation in skeletal muscle
3. Uterine relaxation

Selective β_2 -blockers are used primarily in research rather than clinical practice

- **β_2 = 2 lungs**
- β_2 receptors stimulation result in bronchodilation.





β_1 & β_2 Receptor Agonists
(Non-selective)

Examples: Isoprenaline (Isoproterenol)

- Structurally like epinephrine
- Increases heart rate and contractility
- Relaxation of bronchial, gastrointestinal, and uterine smooth muscle
- Peripheral vasodilation

Uses:

1. Primarily used in the management of brady dysrhythmias

https://www.ncbi.nlm.nih.gov/books/NBK526042/

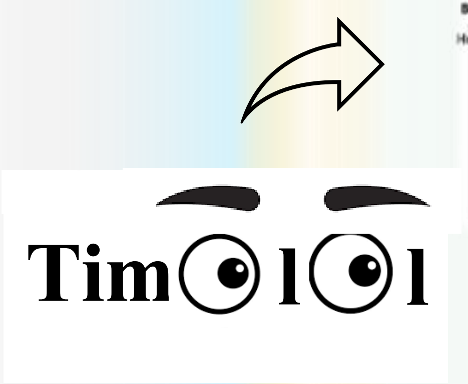
β_1 & β_2 Receptor Antagonists
(Non-selective)

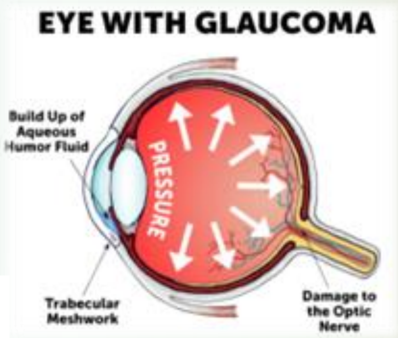
Examples: Propranolol, Timolol, Nadolol

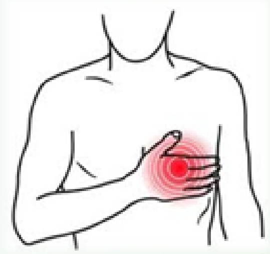
- β-blockers lower BP.
- They are **less likely** than α blockers to induce postural hypotension (α is functional)
- β-mediated bronchodilation is blocked, parasympathetic overactivity will result in bronchospasm

Uses:

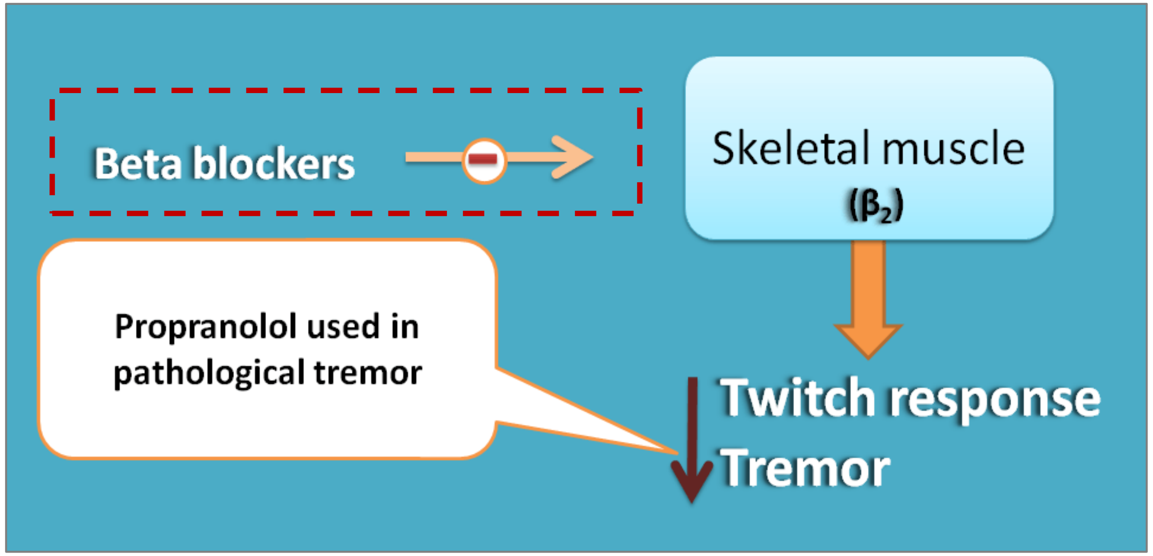
1. Hypertension, angina, arrhythmias
2. Glaucoma (Timolol Aqueous humor production → reduces intraocular pressure)
3. Migraine prophylaxis (Propranolol reduce blood vessel dilation)
4. Propranolol treats tremor & hyperthyroidism (reduces T4 → T3 conversion)

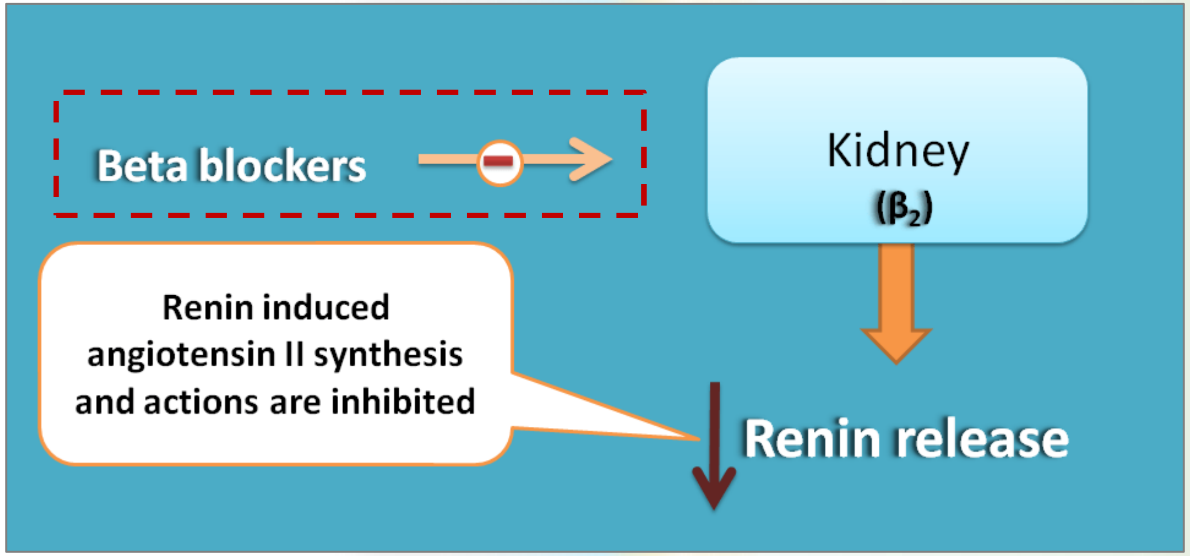






β_1 & β_2 Receptor Antagonists
(Non-selective)





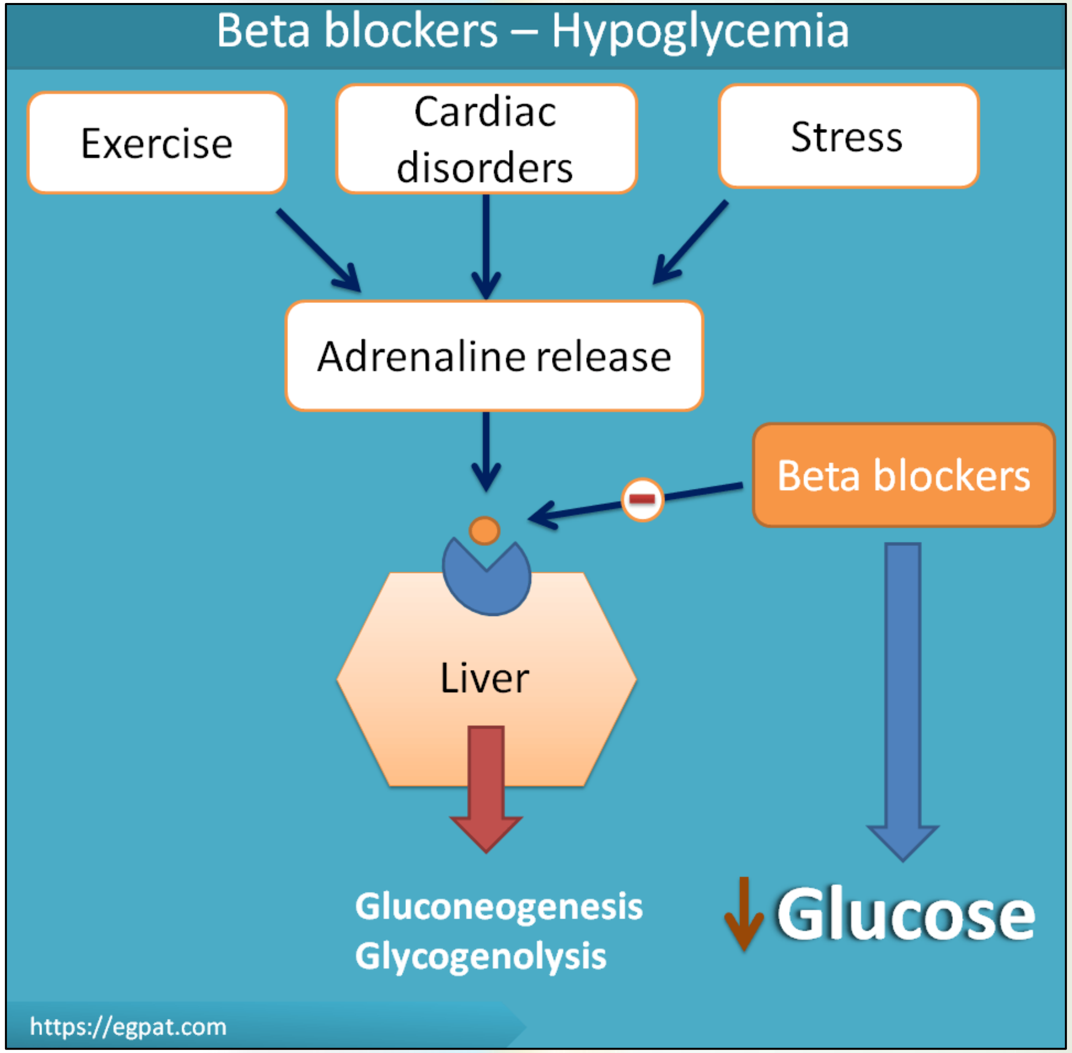
- **β_2 receptors** are predominant on few of the blood vessels supplying their skeletal muscle and liver.
- **β - blockers** decrease blood flow through the skeletal muscle and liver.
- The reduced blood flow to the skeletal muscle decreases twitch responds thereby decrease tremors

- Liver and kidney are supplied by sympathetic system only without any parasympathetic innervation.
- **β - blockers** decreases the renin release by blocking β_1 receptors
- Therefore, activation of renin-angiotensin system and synthesis of angiotensin II are prevented

β_1 & β_2 Receptor Antagonists
(Non-selective)

- **β - blockers** can can mask the symptoms of hypoglycemia (β_2 blockers lose selectivity at high doses).
- **β - blockers** mask crucial warning signs of low blood sugar, such as tachycardia, tremors, and palpitations.
- Sweating (diaphoresis) is usually not masked and may be the only warning sign for a DM patient taking a β - blocker

(be aware of non-adrenergic symptoms of low blood sugar, such as hunger, dizziness, and confusion).



https://www.egpat.com/tutorials/adrenergic-antagonists/actions-uses-of-beta-blockers

β_3 Receptor Agonists

Examples: Mirabegron

- Mirabegron; a β_3 agonist (sympathomimetic) medication
- Used for treating overactive bladder (OAB). It has fewer side effects as compared to the antimuscarinics
- Urinary bladder have sympathetic and parasympathetic receptors to regulate micturition
- The activation of receptors M_2, M_3 result in bladder muscles contraction and micturition (parasympathetic)
- Activation of β3 agonist will cause detrusor muscle relaxation and relief of OAB (sympathetic)

Contraindications:

1. Previous hypersensitivity reaction
2. Severe uncontrolled hypertension

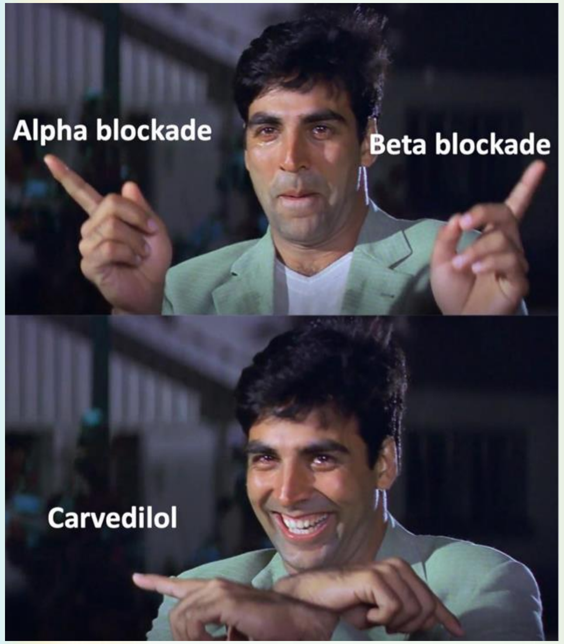
https://www.ncbi.nlm.nih.gov/books/NBK538513/

β_1, β_2 & α_1 Mixed Antagonists

Examples: Labetalol, Carvedilol

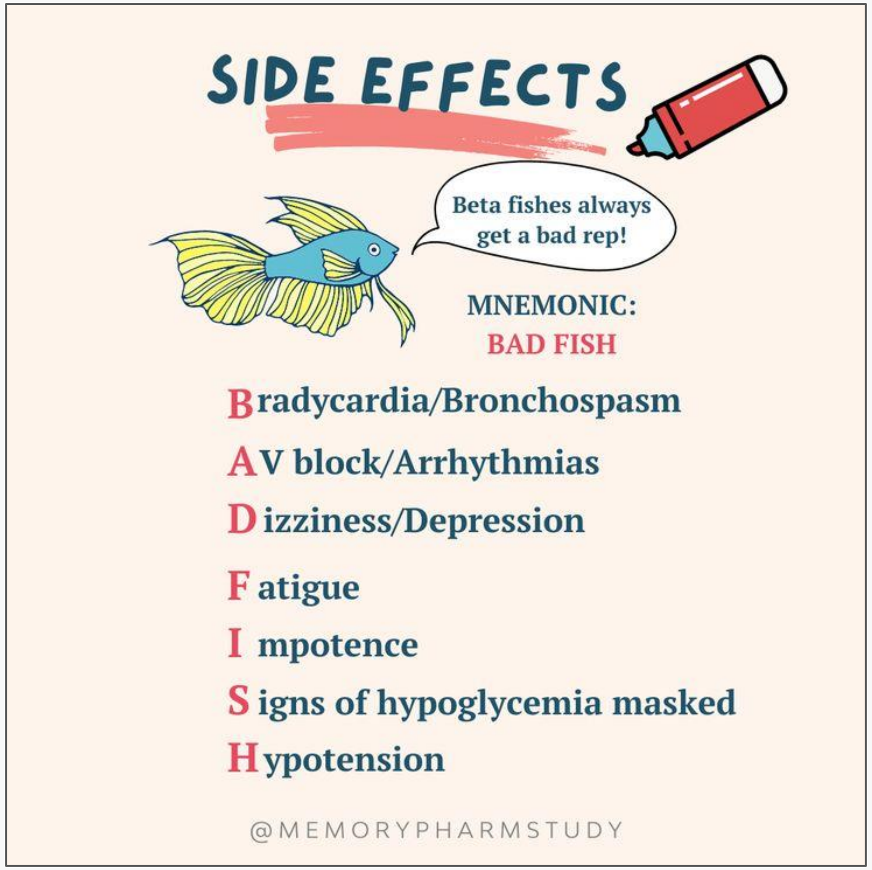
Uses:

1. Hypertension, especially in pregnancy (Labetalol)
2. Heart failure (Carvedilol, helps reduce heart workload)
3. Carvedilol has antioxidant properties, protecting the heart.



β-Receptor Antagonists

(Adverse Effects & contraindications)



Betta Fish (aggressive BAD FISH)

- β-blockers mask hypoglycemia symptoms (tachycardia, tremor).
- They do NOT mask sweating.
- Non-selective β- blockers are contraindicated in asthma

β-Receptor Antagonists

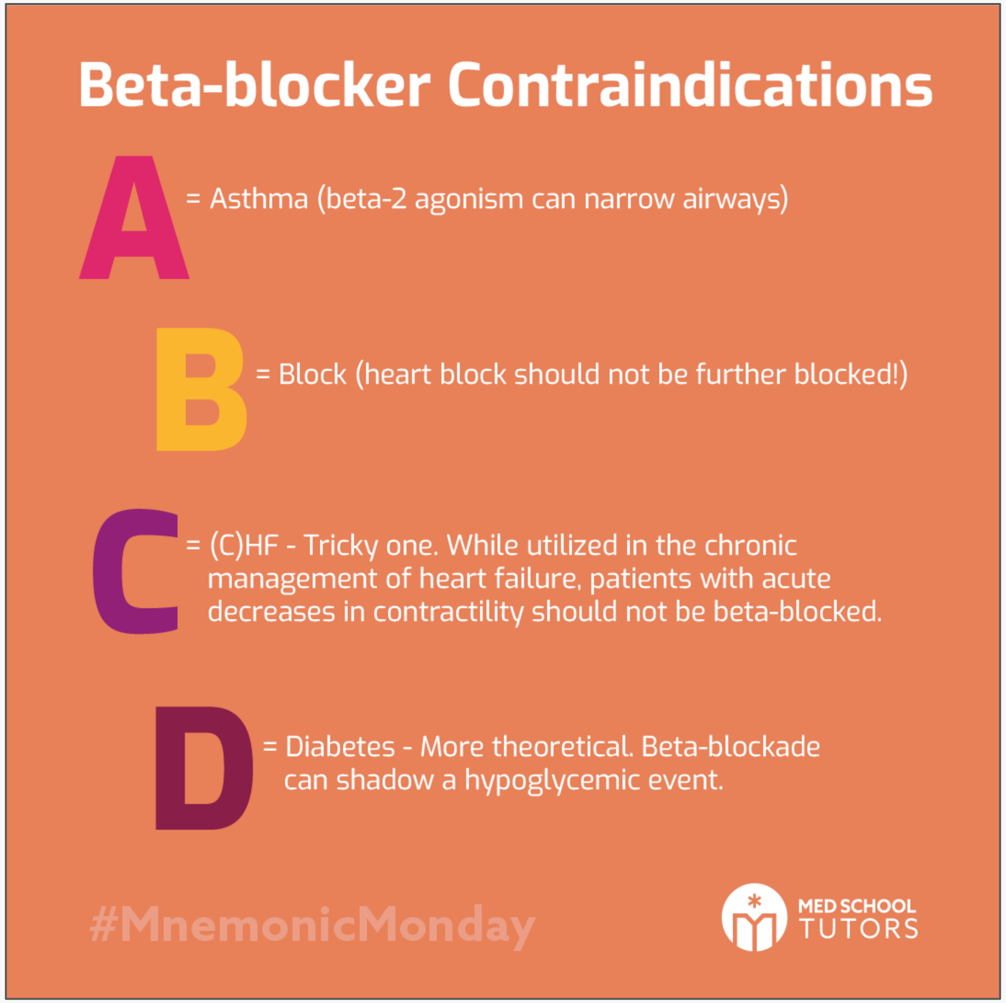
(Adverse Effects & contraindications)

Avoid β-blockers in:

1. Severe asthma
2. AV block
3. Severe bradycardia

Caution in:

1. Diabetes



Keep in Mind!

Type	Examples	Used For	Main Side Effects
$\alpha_1 + \alpha_2$ Blockers (Non-selective)	Phenoxybenzamine, Phentolamine	Pheochromocytoma, Raynaud's disease	Reflex tachycardia, hypotension
α_1 Blockers (Selective)	Prazosin, Doxazosin, Tamsulosin	Hypertension, BPH	First-dose syncope, dizziness
$\beta_1 + \beta_2$ Blockers (Non-selective)	Propranolol, Timolol	Hypertension, arrhythmias, migraine, glaucoma	Bronchoconstriction, fatigue
β_1 Blockers (Selective)	Atenolol, Metoprolol, Esmolol	Hypertension, heart failure, arrhythmias	Bradycardia, hypotension
$\beta + \alpha_1$ Blockers	Labetalol, Carvedilol	Hypertension, heart failure	Dizziness, fatigue

Don’t Forget!

Examples	Used For
Sweat glands	Sympathetic BUT M3
Heart parasympathetic receptor	M2
Atropine	Blocks M2 → ↑ HR. Blocks M3 → dry mouth, ↓ sweating
Vascular tone	Mainly α_1
Endothelial vasodilation	M3 → NO

One More Time!

Examples	Used For
Epinephrine	Emergency drug (anaphylaxis, cardiac arrest)
Norepinephrine	Pressure drug (septic shock $\rightarrow$ increase BP safely)
Dopamine	Limited clinical use (arrhythmia risk + no renal benefit)
Epinephrine at low dose	Drop diastolic BP (β_2 vasodilation)
Norepinephrine	Cause bradycardia despite β_1 activity (reflex)
Labetalol, Carvedilol	Hypertension, heart failure

References

1. Lippincott Illustrated Reviews Pharmacology

2. www.Amboss.com

3. Katzung Basic & Clinical Pharmacology